Operating System Security

Data Recovery using Hiren’s BootCD PE

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QUESTION

Recover a music file deleted from any memory stick using Hiren’s BootCD PE.

Introduction

- ✓In the world of modern computing, data loss can be catastrophic.
- ✓A simple deletion, whether accidental or intentional, can seem like the end of a file.
- ✓However, with the right tools and knowledge, files can be recovered.
- ✓In this presentation, we will demonstrate how to recover a deleted music file using **Hiren's BootCD PE.**

Objectives

- ✓To demonstrate the recovery of a deleted music file using Hiren’s BootCD PE (Recuva).
- ✓To understand the file recovery process within an operating system environment.
- ✓To explore the capabilities of Hiren’s BootCD PE in data restoration and OS security.
- ✓To emphasize the role of data recovery tools in mitigating data loss risks.

Hiren’s BootCD PE

- ✓Download **Hiren’s BootCD PE**
- ✓**Hiren's Boot CD** is an all-in-one bootable rescue disc solution that contains a load of useful Windows repair tools you'll want to have when you can no longer boot into your PC.
- ✓Imagine issues like **hard drive failure**, **virus infections**, **partitioning**, **password recovery**, and **data recovery**.
- ✓Source → **Make use of** (website)

Data Recovery

- ✓**Data recovery** is the process of retrieving lost, deleted, damaged, or inaccessible data from storage devices, often involving specialized software and techniques to scan, identify, and restore the data.
- ✓In digital forensics, a wide array of data can be recovered, including **deleted files**, **emails**, **browsing history**, **metadata**, and even data from encrypted or damaged devices.
- ✓Our focus in on **deleted files (music).**

How to Recover Deleted Files Using Hiren’s BootCD PE

Creating a Bootable Device

1. If you've downloaded Hiren's BootCD PE from the actual web, the file will be written as .iso
2. Use a booting tool such as Rufus or PowerISO to make the .iso file bootable into your USB drive.

OR

1. If you have the Hiren's BootCD PE as a folder, copy the files into an USB drive and you are ready to go.

Initializing and Loading Up Hiren's BootCD PE

1. Shutdown and restart your PC into the Boot menu by pressing F12 repeatedly depending on your manufacturer.
2. Select the bootable USB drive e.g., UEFI: VendorCoProductCode 2.00, Partition 1
3. Wait for Hiren's BootCD PE to initialize and load up.

Data Recover With Hirens BootCD PE

Steps

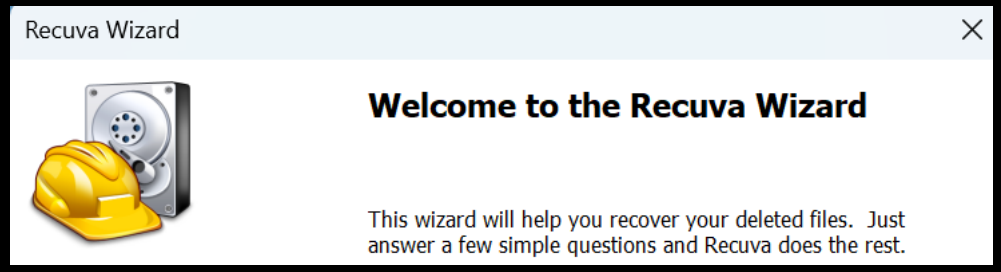
1. Navigate to the folder called **Utilities** (Located on the Desktop)
2. Navigate to the folder called **Hard Disk Tools**
3. Navigate to the folder called **Data Recovery**

Tools

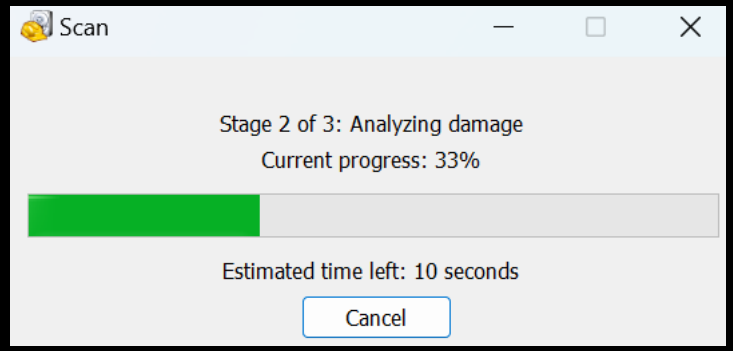
1. AppleFS for Windows
2. Check Disk GUI
3. DME
4. Lazesoft Data Recovery
5. Linux Reader
6. PhotoRec
7. Puran Data Recovery
8. ReclaimMe Free RAID Recovery
9. Recuva
10. Unstoppable Copier
11. Victoria

Recuva

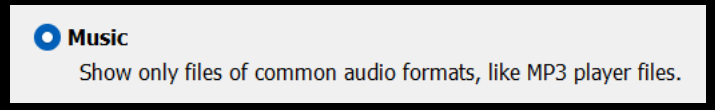
✓Open Recuva



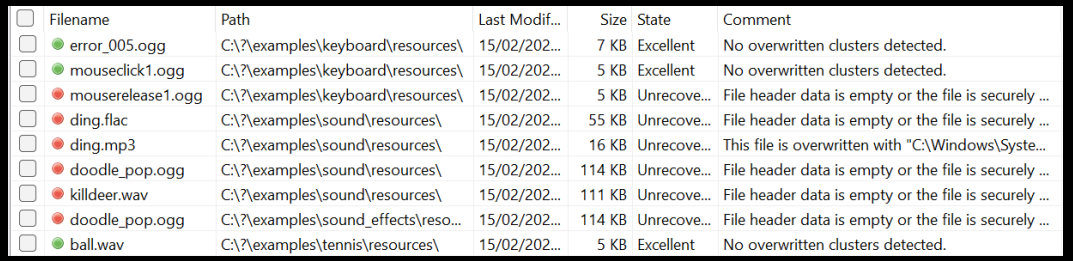
✓Start the Scan



✓Select the File Type

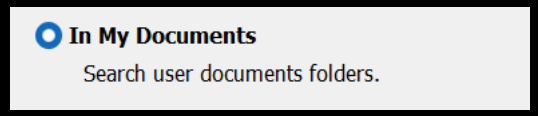


✓Switch to Advanced Mode



Filename	Path	Last Modif...	Size	State	Comment
error_005.ogg	C:\?\examples\keyboard\resources\	15/02/202...	7 KB	Excellent	No overwritten clusters detected.
mouseclick1.ogg	C:\?\examples\keyboard\resources\	15/02/202...	5 KB	Excellent	No overwritten clusters detected.
mouserelease1.ogg	C:\?\examples\keyboard\resources\	15/02/202...	5 KB	Unrecove...	File header data is empty or the file is securely ...
ding.flac	C:\?\examples\sound\resources\	15/02/202...	55 KB	Unrecove...	File header data is empty or the file is securely ...
ding.mp3	C:\?\examples\sound\resources\	15/02/202...	16 KB	Unrecove...	This file is overwritten with "C:\Windows\Syste...
doodle_pop.ogg	C:\?\examples\sound\resources\	15/02/202...	114 KB	Unrecove...	File header data is empty or the file is securely ...
killdeer.wav	C:\?\examples\sound\resources\	15/02/202...	111 KB	Unrecove...	File header data is empty or the file is securely ...
doodle_pop.ogg	C:\?\examples\sound_effects\reso...	15/02/202...	114 KB	Unrecove...	File header data is empty or the file is securely ...
ball.wav	C:\?\examples\tennis\resources\	15/02/202...	5 KB	Excellent	No overwritten clusters detected.

✓Select the File Location



The Role of Data Recovery Tools in Mitigating Data Loss Risks

1. **File Recovery** – They provide the ability to recover lost data from various storage mediums, ensuring important files are not permanently lost due to accidental deletion or system failures.

2. **Minimizing Downtime** – These tools enable quick restoration of data, reducing downtime and maintaining business continuity in both personal and professional environments.

3. **Data Persistence** – By recovering seemingly unrecoverable files, data recovery tools highlight the importance of data persistence and ensure that critical information is safeguarded.

Let’s Jump Into the Practical

End of The Theory Presentation